Newman-Penrose Formalism Calculations

Stephani et al., Equation (16.75)

Metric

$$ds^2 = -\left(ar^2 + bt\right)^{\frac{4}{3}} \mathrm{d}t \otimes \mathrm{d}t + \left(ar^2 + bt\right)^{\frac{2}{3}} \mathrm{d}r \otimes \mathrm{d}r + \left(ar^2 + bt\right)^{\frac{2}{3}} r^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + \left(ar^2 + bt\right)^{\frac{2}{3}} r^2 \sin\left(\theta\right)^2 \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Metric Functions

$$A = \left(ar^2 + bt\right)^{\frac{2}{3}}$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\left(ar^2 + bt\right)^{\frac{2}{3}}\right) dt + \left(\sqrt{\frac{1}{2}}\left(ar^2 + bt\right)^{\frac{1}{3}}\right) dr$$

$$n = \left(\frac{1}{2}\sqrt{2}\left(ar^2 + bt\right)^{\frac{2}{3}}\right) dt + \left(-\sqrt{\frac{1}{2}}\left(ar^2 + bt\right)^{\frac{1}{3}}\right) dr$$

$$m = \left(\frac{1}{2}\sqrt{2}\left(ar^2 + bt\right)^{\frac{1}{3}} r\right) d\theta + \left(\frac{1}{2} i \sqrt{2}\left(ar^2 + bt\right)^{\frac{1}{3}} r \sin\left(\theta\right)\right) d\phi$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}\left(ar^2 + bt\right)^{\frac{1}{3}} r\right) d\theta + \left(-\frac{1}{2} i \sqrt{2}\left(ar^2 + bt\right)^{\frac{1}{3}} r \sin\left(\theta\right)\right) d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{2}{3}}}\right)\,dt + \left(\frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}}\right)\,dr$$

$$n = \left(-\frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{2}{3}}}\right)\,dt + \left(-\frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}}\right)\,dr$$

$$m = \left(\frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}r}\right)\,d\theta + \left(\frac{i\,\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}r\sin\left(\theta\right)}\right)\,d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}r}\right)\,d\theta + \left(-\frac{i\,\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}r\sin\left(\theta\right)}\right)\,d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{2}{3}}}\,\partial_t + \frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}}\,\partial_r$$

$$\Delta = -\frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{2}{3}}}\,\partial_t - \frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}}\,\partial_r$$

$$\delta = \frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}r}\,\partial_\theta + \frac{i\,\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}r\sin\left(\theta\right)}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}r}\,\partial_\theta - \frac{i\,\sqrt{2}}{2\left(ar^2+bt\right)^{\frac{1}{3}}r\sin\left(\theta\right)}\,\partial_\phi$$

Spin Coefficients

$$\rho = -\frac{5\,\sqrt{2}a^{2}r^{4}}{6\,(a^{2}r^{5} + 2\,abr^{3}t + b^{2}rt^{2})(ar^{2} + bt)^{\frac{1}{3}}} - \frac{4\,\sqrt{2}abr^{2}t}{3\,(a^{2}r^{5} + 2\,abr^{3}t + b^{2}rt^{2})(ar^{2} + bt)^{\frac{1}{3}}} - \frac{\sqrt{2}b^{2}t^{2}}{2\,(a^{2}r^{5} + 2\,abr^{3}t + b^{2}rt^{2})(ar^{2} + bt)^{\frac{1}{3}}} + \frac{\sqrt{2}\left(ar^{2} + bt\right)^{\frac{1}{3}}br}{6\,(a^{2}r^{5} + 2\,abr^{3}t + b^{2}rt^{2})}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{5\,\sqrt{2}a^{2}r^{4}}{6\,(a^{2}r^{5} + 2\,abr^{3}t + b^{2}rt^{2})(ar^{2} + bt)^{\frac{1}{3}}} - \frac{4\,\sqrt{2}abr^{2}t}{3\,(a^{2}r^{5} + 2\,abr^{3}t + b^{2}rt^{2})(ar^{2} + bt)^{\frac{1}{3}}} - \frac{\sqrt{2}b^{2}t^{2}}{2\,(a^{2}r^{5} + 2\,abr^{3}t + b^{2}rt^{2})(ar^{2} + bt)^{\frac{1}{3}}} - \frac{\sqrt{2}\left(ar^{2} + bt\right)^{\frac{1}{3}}br}{6\,(a^{2}r^{5} + 2\,abr^{3}t + b^{2}rt^{2})}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\cos\left(\theta\right)}{4\,(ar^{2} + bt)^{\frac{1}{3}}r\sin\left(\theta\right)}$$

$$\beta = \frac{\sqrt{2}\cos\left(\theta\right)}{4\,(ar^{2} + bt)^{\frac{1}{3}}r\sin\left(\theta\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2}\left(ar^{2} + bt\right)^{\frac{2}{3}}ar}{3\,(a^{2}r^{4} + 2\,abr^{2}t + b^{2}t^{2})} - \frac{\sqrt{2}\left(ar^{2} + bt\right)^{\frac{1}{3}}b}{12\,(a^{2}r^{4} + 2\,abr^{2}t + b^{2}t^{2})}$$

$$\gamma = \frac{\sqrt{2}\left(ar^2 + bt\right)^{\frac{2}{3}} ar}{3\,(a^2r^4 + 2\,abr^2t + b^2t^2)} + \frac{\sqrt{2}\left(ar^2 + bt\right)^{\frac{1}{3}} b}{12\,(a^2r^4 + 2\,abr^2t + b^2t^2)}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{10\left(ar^2 + bt\right)^{\frac{1}{3}} a^3r^4}{9\,(a^4r^8 + 4\,a^3br^6t + 6\,a^2b^2r^4t^2 + 4\,ab^3r^2t^3 + b^4t^4)} + \frac{10\left(ar^2 + bt\right)^{\frac{1}{3}} a^2br^2t}{9\,(a^4r^8 + 4\,a^3br^6t + 6\,a^2b^2r^4t^2 + 4\,ab^3r^2t^3 + b^4t^4)} - \frac{10\,a^2br^3}{9\,(a^4r^8 + 4\,a^3br^6t + 6\,a^2b^2r^4t^2 + 4\,ab^3r^2t^3 + b^4t^4)} - \frac{10\,ab^2rt}{9\,(a^4r^8 + 4\,a^3br^6t + 6\,a^2b^2r^4t^2 + 4\,ab^3r^2t^3 + b^4t^4)} + \frac{5\left(ar^2 + bt\right)^{\frac{2}{3}} b^2}{18\,(a^4r^8 + 4\,a^3br^6t + 6\,a^2b^2r^4t^2 + 4\,ab^3r^2t^3 + b^4t^4)}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{5\left(ar^2 + bt\right)^{\frac{1}{3}} a^2r^2}{9\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)} + \frac{5\,b^2}{36\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)(ar^2 + bt)^{\frac{1}{3}}}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{10\left(ar^2 + bt\right)^{\frac{1}{3}} a^3r^4}{9\,(a^4r^8 + 4\,a^3br^6t + 6\,a^2b^2r^4t^2 + 4\,ab^3r^2t^3 + b^4t^4)} + \frac{10\left(ar^2 + bt\right)^{\frac{1}{3}} a^2br^2t}{9\,(a^4r^8 + 4\,a^3br^6t + 6\,a^2b^2r^4t^2 + 4\,ab^3r^2t^3 + b^4t^4)} + \frac{10\,a^2br^3}{9\,(a^4r^8 + 4\,a^3br^6t + 6\,a^2b^2r^4t^2 + 4\,ab^3r^2t^3 + b^4t^4)} + \frac{10\,ab^2rt}{9\,(a^4r^8 + 4\,a^3br^6t + 6\,a^2b^2r^4t^2 + 4\,ab^3r^2t^3 + b^4t^4)} + \frac{5\left(ar^2 + bt\right)^{\frac{2}{3}} b^2}{18\,(a^4r^8 + 4\,a^3br^6t + 6\,a^2b^2r^4t^2 + 4\,ab^3r^2t^3 + b^4t^4)}$$

$$\Lambda = -\frac{13\,a^3r^4}{27\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)(ar^2 + bt)^{\frac{2}{3}}} - \frac{31\,a^2br^2t}{27\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)(ar^2 + bt)^{\frac{2}{3}}} - \frac{2\,ab^2t^2}{3\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)(ar^2 + bt)^{\frac{2}{3}}} - \frac{b^2}{12\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)(ar^2 + bt)^{\frac{1}{3}}}$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$\Psi_2 = -\frac{4\,a^2r^2}{27\,(a^2r^4 + 2\,abr^2t + b^2t^2)(ar^2 + bt)^{\frac{2}{3}}}$$

$\Psi_3 = 0$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"